

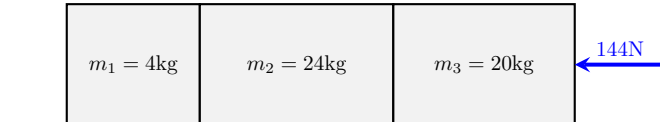
Dynamics Test Answers

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Blocks

1. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 144N force acts on block 3 from the right, and the surface can be considered frictionless. Calculate:



- (a) The acceleration of the system

Solution: The acceleration of the system is 3 ms^{-2} left

- (b) The acceleration of each block

Solution: Each block has an acceleration of 3 ms^{-2} left

- (c) The net force on each block

Solution: Block 1 has a net force of 12N left

Block 2 has a net force of 72N left

Block 3 has a net force of 60N left

- (d) The contact forces for each block

Solution: Block 1 has 1 force acting on it. They are:

12N left Contact Force from block 2

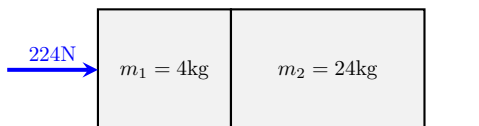
Block 2 has 2 forces acting on it. They are:

12N right Contact Force from block 1, 84N left Contact Force from block 3

Block 3 has 2 forces acting on it. They are:

144N left Applied Force, 84N right Contact Force from block 2

2. (6 points) The diagram shows 2 blocks in contact on a smooth surface. A 224N force acts on block 1 from the left, and the surface can be considered frictionless. Calculate:



- (a) The acceleration of the system

Solution: The acceleration of the system is 8 ms^{-2} right

- (b) The acceleration of each block

Solution: Each block has an acceleration of 8 ms^{-2} right

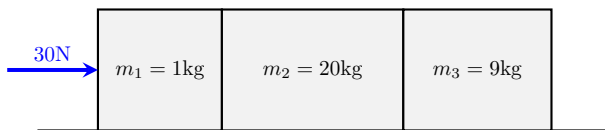
- (c) The net force on each block

Solution: Block 1 has a net force of 32N right
Block 2 has a net force of 192N right

- (d) The contact forces for each block

Solution: Block 1 has 2 forces acting on it. They are:
224N right Applied Force, 192N left Contact Force from block 2
Block 2 has 1 force acting on it. They are:
192N right Contact Force from block 1

3. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 30N force acts on block 1 from the left, and the surface can be considered frictionless. Calculate:



- (a) The acceleration of the system

Solution: The acceleration of the system is 1 ms^{-2} right

- (b) The acceleration of each block

Solution: Each block has an acceleration of 1 ms^{-2} right

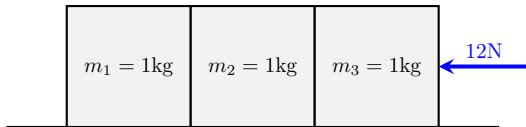
- (c) The net force on each block

Solution: Block 1 has a net force of 1N right
Block 2 has a net force of 20N right
Block 3 has a net force of 9N right

- (d) The contact forces for each block

Solution: Block 1 has 2 forces acting on it. They are:
30N right Applied Force, 29N left Contact Force from block 2
Block 2 has 2 forces acting on it. They are:
29N right Contact Force from block 1, 9N left Contact Force from block 3
Block 3 has 1 force acting on it. They are:
9N right Contact Force from block 2

4. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 12N force acts on block 3 from the right, and the surface can be considered frictionless. Calculate:



- (a) The acceleration of the system

Solution: The acceleration of the system is 4 ms^{-2} left

- (b) The acceleration of each block

Solution: Each block has an acceleration of 4 ms^{-2} left

- (c) The net force on each block

Solution: Block 1 has a net force of 4N left

Block 2 has a net force of 4N left

Block 3 has a net force of 4N left

- (d) The contact forces for each block

Solution: Block 1 has 1 force acting on it. They are:

4N left Contact Force from block 2

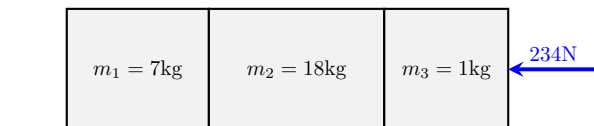
Block 2 has 2 forces acting on it. They are:

4N right Contact Force from block 1, 8N left Contact Force from block 3

Block 3 has 2 forces acting on it. They are:

12N left Applied Force, 8N right Contact Force from block 2

5. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 234N force acts on block 3 from the right, and the surface can be considered frictionless. Calculate:



- (a) The acceleration of the system

Solution: The acceleration of the system is 9 ms^{-2} left

- (b) The acceleration of each block

Solution: Each block has an acceleration of 9 ms^{-2} left

- (c) The net force on each block

Solution: Block 1 has a net force of 63N left

Block 2 has a net force of 162N left

Block 3 has a net force of 9N left

- (d) The contact forces for each block

Solution: Block 1 has 1 force acting on it. They are:

63N left Contact Force from block 2

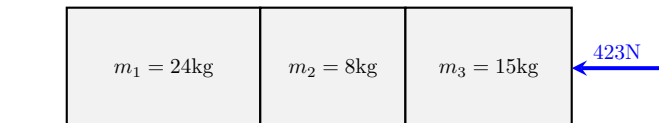
Block 2 has 2 forces acting on it. They are:

63N right Contact Force from block 1, 225N left Contact Force from block 3

Block 3 has 2 forces acting on it. They are:

234N left Applied Force, 225N right Contact Force from block 2

6. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 423N force acts on block 3 from the right, and the surface can be considered frictionless. Calculate:



- (a) The acceleration of the system

Solution: The acceleration of the system is 9 ms^{-2} left

- (b) The acceleration of each block

Solution: Each block has an acceleration of 9 ms^{-2} left

- (c) The net force on each block

Solution: Block 1 has a net force of 216N left

Block 2 has a net force of 72N left

Block 3 has a net force of 135N left

- (d) The contact forces for each block

Solution: Block 1 has 1 force acting on it. They are:

216N left Contact Force from block 2

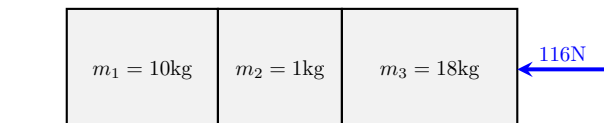
Block 2 has 2 forces acting on it. They are:

216N right Contact Force from block 1, 288N left Contact Force from block 3

Block 3 has 2 forces acting on it. They are:

423N left Applied Force, 288N right Contact Force from block 2

7. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 116N force acts on block 3 from the right, and the surface can be considered frictionless. Calculate:



- (a) The acceleration of the system

Solution: The acceleration of the system is 4 ms^{-2} left

- (b) The acceleration of each block

Solution: Each block has an acceleration of 4 ms^{-2} left

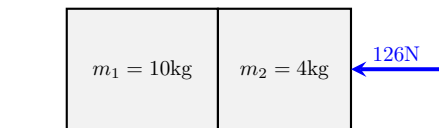
- (c) The net force on each block

Solution: Block 1 has a net force of 40N left
Block 2 has a net force of 4N left
Block 3 has a net force of 72N left

- (d) The contact forces for each block

Solution: Block 1 has 1 force acting on it. They are:
40N left Contact Force from block 2
Block 2 has 2 forces acting on it. They are:
40N right Contact Force from block 1, 44N left Contact Force from block 3
Block 3 has 2 forces acting on it. They are:
116N left Applied Force, 44N right Contact Force from block 2

8. (6 points) The diagram shows 2 blocks in contact on a smooth surface. A 126N force acts on block 2 from the right, and the surface can be considered frictionless. Calculate:



- (a) The acceleration of the system

Solution: The acceleration of the system is 9 ms^{-2} left

- (b) The acceleration of each block

Solution: Each block has an acceleration of 9 ms^{-2} left

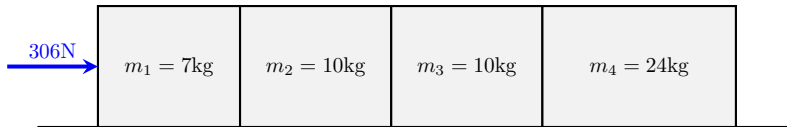
- (c) The net force on each block

Solution: Block 1 has a net force of 90N left
Block 2 has a net force of 36N left

- (d) The contact forces for each block

Solution: Block 1 has 1 force acting on it. They are:
90N left Contact Force from block 2
Block 2 has 2 forces acting on it. They are:
126N left Applied Force, 90N right Contact Force from block 1

9. (6 points) The diagram shows 4 blocks in contact on a smooth surface. A 306N force acts on block 1 from the left, and the surface can be considered frictionless. Calculate:



- (a) The acceleration of the system

Solution: The acceleration of the system is 6 ms^{-2} right

- (b) The acceleration of each block

Solution: Each block has an acceleration of 6 ms^{-2} right

- (c) The net force on each block

Solution: Block 1 has a net force of 42N right

Block 2 has a net force of 60N right

Block 3 has a net force of 60N right

Block 4 has a net force of 144N right

- (d) The contact forces for each block

Solution: Block 1 has 2 forces acting on it. They are:

306N right Applied Force, 264N left Contact Force from block 2

Block 2 has 2 forces acting on it. They are:

264N right Contact Force from block 1, 204N left Contact Force from block 3

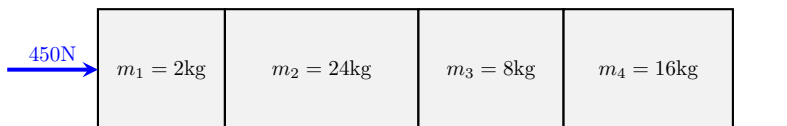
Block 3 has 2 forces acting on it. They are:

204N right Contact Force from block 2, 144N left Contact Force from block 4

Block 4 has 1 force acting on it. They are:

144N right Contact Force from block 3

10. (6 points) The diagram shows 4 blocks in contact on a smooth surface. A 450N force acts on block 1 from the left, and the surface can be considered frictionless. Calculate:



- (a) The acceleration of the system

Solution: The acceleration of the system is 9 ms^{-2} right

- (b) The acceleration of each block

Solution: Each block has an acceleration of 9 ms^{-2} right

- (c) The net force on each block

Solution: Block 1 has a net force of 18N right
Block 2 has a net force of 216N right
Block 3 has a net force of 72N right
Block 4 has a net force of 144N right

(d) The contact forces for each block

Solution: Block 1 has 2 forces acting on it. They are:
450N right Applied Force, 432N left Contact Force from block 2
Block 2 has 2 forces acting on it. They are:
432N right Contact Force from block 1, 216N left Contact Force from block 3
Block 3 has 2 forces acting on it. They are:
216N right Contact Force from block 2, 144N left Contact Force from block 4
Block 4 has 1 force acting on it. They are:
144N right Contact Force from block 3